

TIME SERIES ANALYSIS

PROJECT:

RAINFALL IN TWO DIFFERENT REGIONS OF INDIA



AIM:

To calculate the components of a time series data and hence find their significance to forecast the rainfall patterns till the year 2025.

PROBLEM STATEMENT:

Comparative analysis of rainfall patterns in two regions of India: Delhi Haryana & Chandigarh (region we reside in) and Assam and Meghalaya (region having the highest rainfall in India) by measuring and isolating each component and identifying the factors responsible for them and to conduct Box Jenkins analysis to identify ,estimate ,check and finally produce forecast from an appropriate model for both the time series data.

Link to the dataset:

https://docs.google.com/spreadsheets/d/1hOh5VOtqasFUVZuL2lowH-J1T67xVq_0gAQxstg258E/edit#gid=849064342

TIME SERIES - AN INTRODUCTION

- Time series is a set of numerical data observed at different points of time, giving information about the past behaviour of a phenomenon or variable of interest. The analysis of time series is of great significance not only to the economists and businessmen but also to the scientists, astronomers, geologists, sociologists, biologists, research worker etc.
- Usually, we shall come across time series showing continual changes over time, giving the overall impression of a haphazard movement.
- The part which can be accounted for and can be attributed specific reasons is the systematic part and the remaining part is unsystematic or irregular.
 - > The types of factors or movements characterizing a time series are called components of a time series.
- Almost all the time series contain unsystematic or irregular part.
- The systematic part is attributed to factors
 - (i) Secular trend (ii) seasonal variation, (iii) cyclical variation.
- > A multiplicative model has been used to explain rainfall, since all components are interdependent.

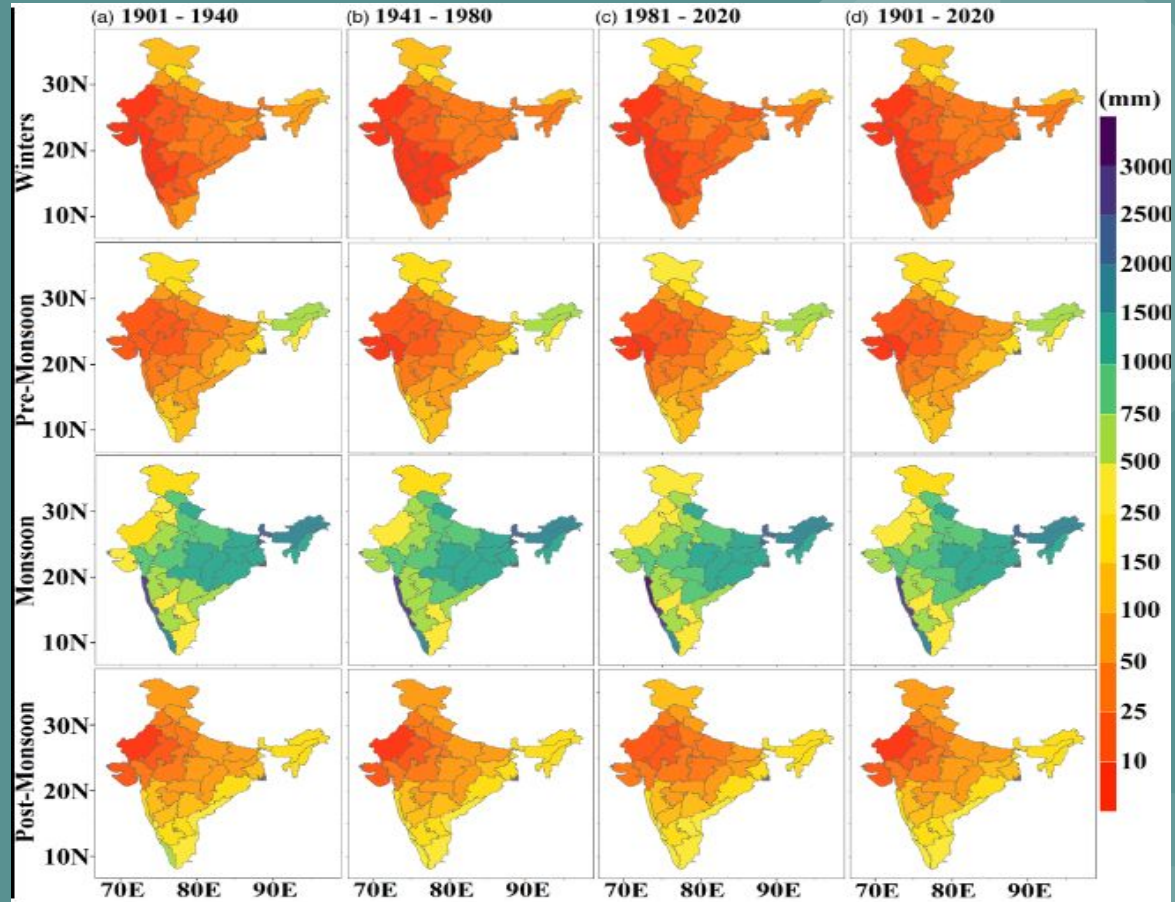
$$Y_t = T_t S_t R_t C_t$$

Where T_t represents trend, S_t represents seasonal component, R_t represents random component and C_t represents cyclic component

Rainfall patterns from 1966-2015

Haryana is popularly known as the breadbasket of India as it provides significant amounts of paddy during kharif season and wheat during rabi season. Assam-Meghalaya receives the highest rainfall of the country after Arunachal Pradesh. These regions are vital to India's agricultural landscape, making their rainfall patterns of utmost importance for food security and the overall economy. Climate change and climate variability significantly influence the most important component of global hydrological cycle, i.e., rainfall. Thus it is very important to look into the rainfall patterns in both these regions.

RAINFALL PATTERN IN VARIOUS REGIONS OF INDIA OVER YEARS



DELHI - HARYANA

Among different seasons, monsoon season had received maximum rain days, followed by pre-monsoon, winter, and post-monsoon season. The number of rain days has decreased during the winter and post-monsoon seasons, while increased during the pre-monsoon season in most of the districts of Haryana as per the innovative trend analysis method (ITAM). In most districts, the monsoon season has exhibited a declining trend in moderate intensity whereas increasing trend in both light and heavy rainfall intensity category rainfall events.

Understanding the rainfall dynamics helps in optimize agricultural or other socio-economic activities such as diversification of crops based on onset and amount of monsoon rainfall, adaptation against droughts and floods etc.

Rainfall % Departures from the Long Period Averages for Districts in HARYANA

PERIOD : 01.06.2020 - 30.06.2020



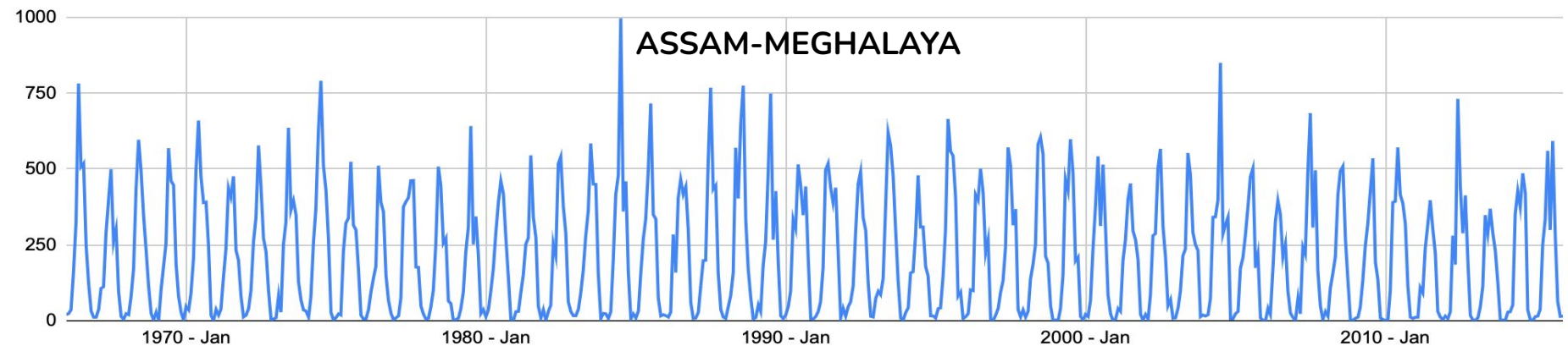
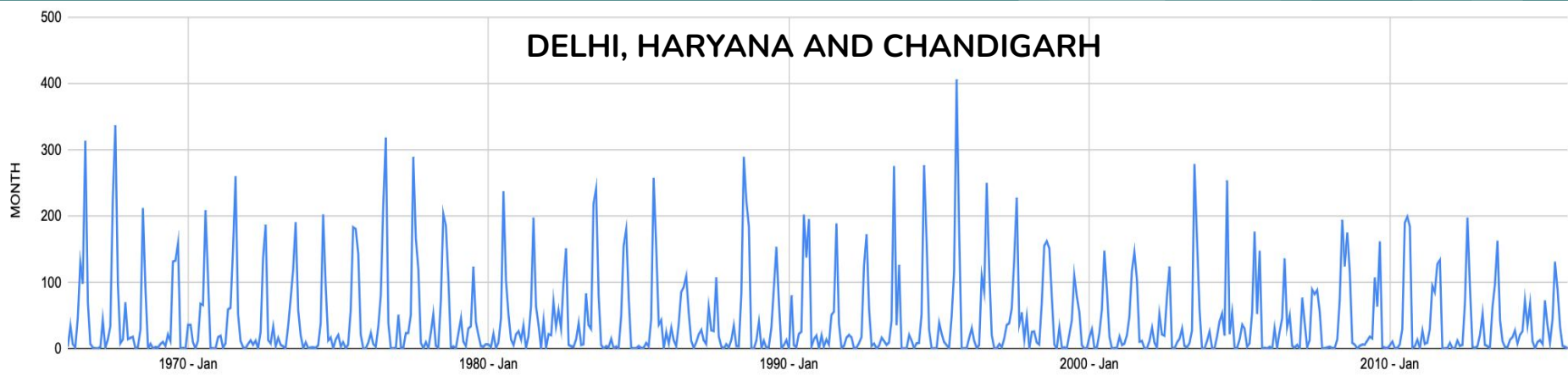
Climate change impacts rainfall distribution in northeast; results in prolonged dry days

GUWAHATI: Climate change has impacted [rainfall](#) distribution during monsoon months in northeast even as the total amount of rain remains mostly unchanged. On the eve of World Environment Day, experts said this lack of uniform distribution of rainfall throughout the monsoon months has resulted in prolonged consecutive dry days and shorter wet days.

Research on rainfall patterns at Assam Agricultural University (Aau) revealed a decreasing tendency of annual rainfall by 66.4 mm/decade during 1986-2015 over the Brahmaputra valley of Assam.

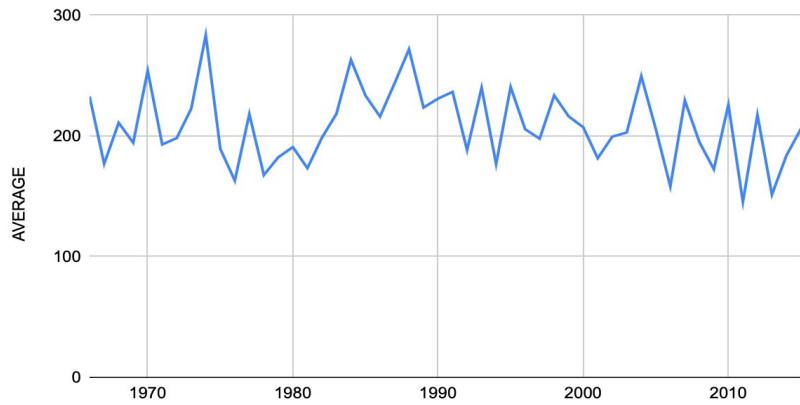
An article by Times of
India, Jun 4, 2023,
21:50 IST

TIME PLOT

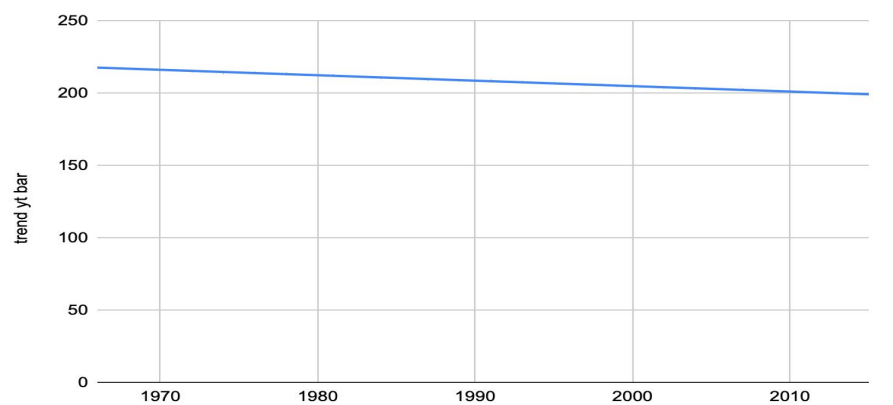


TREND-Assam and Meghalaya

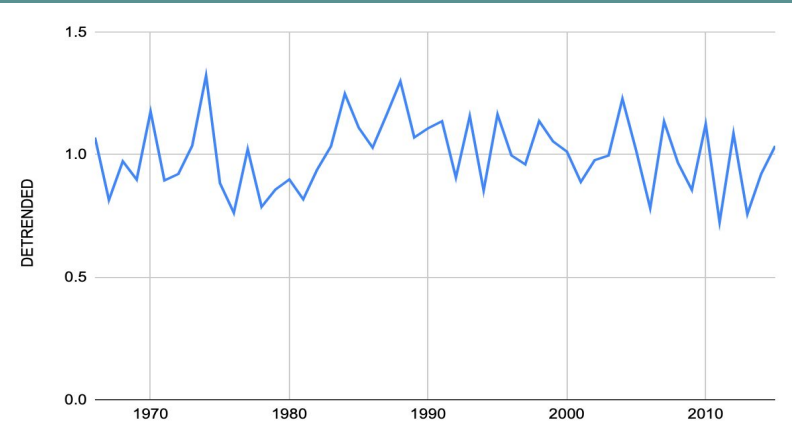
TIME PLOT



TREND



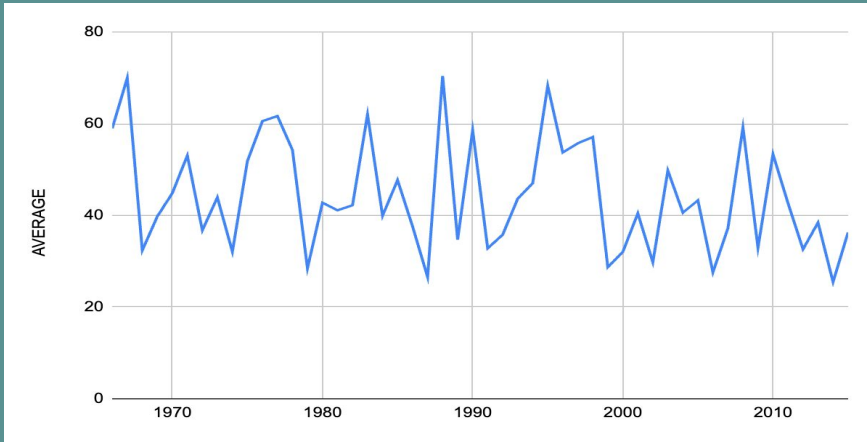
DETRENNED



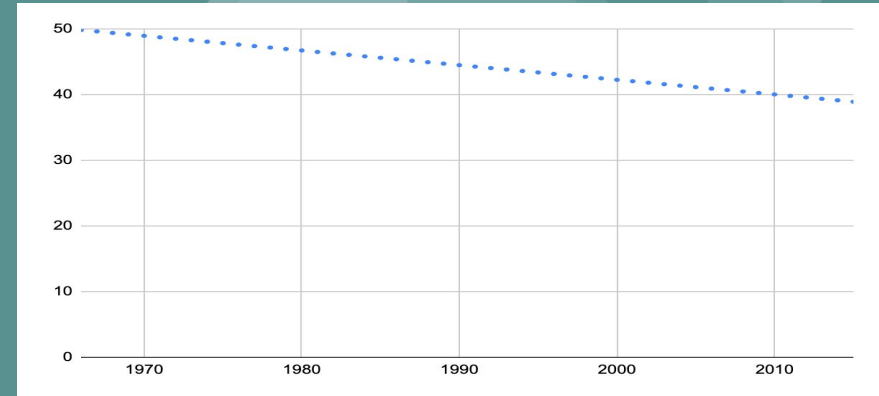
1. A linear trend is fitted on the average rainfall in each year.
2. We can observe a little decline in linear trend over a long period of time.
3. Since the trend present in the data is very minimal, the detrended plot looks very similar to the original time plot.
4. The trend may be attributed to historical evidence of climate change, deforestation, urbanization, and natural climate variability.

TREND- Delhi Haryana and Chandigarh

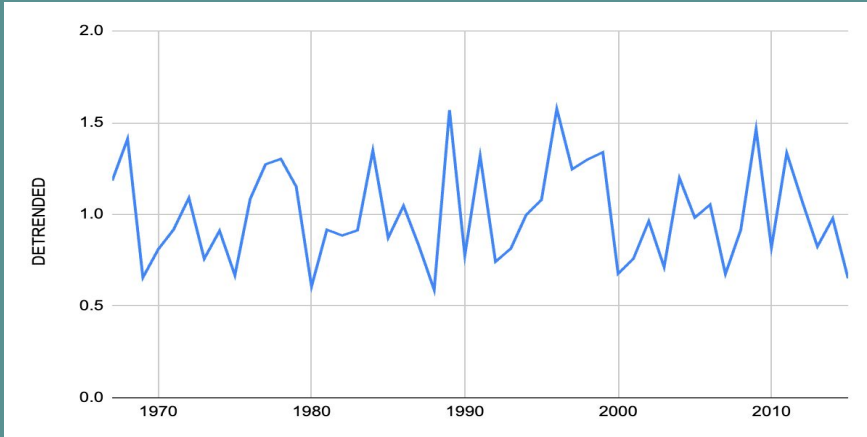
TIME PLOT



TREND



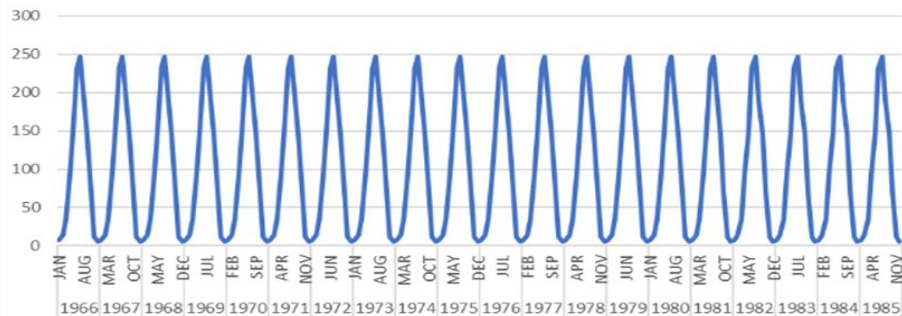
DETRENDED



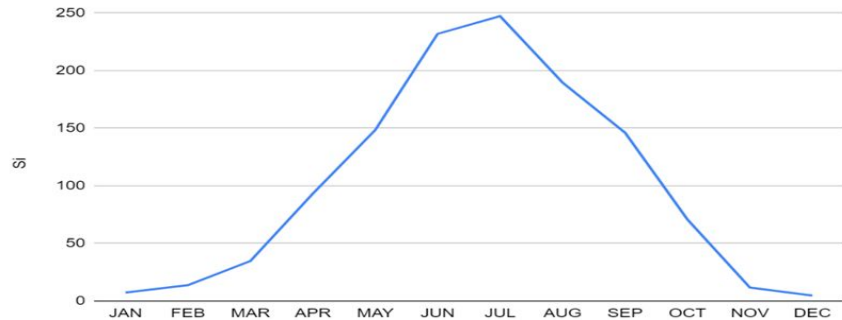
1. A linear trend is fitted on the average rainfall in each year.
2. We can observe a decline in linear trend over a long period of time.
3. The trend is present in the data, the detrended plot varies only a little bit from the original time plot.
4. The trend may be attributed to historical evidence of climate change, deforestation, urbanization, and natural climate variability.

SEASONAL COMPONENT-Assam Meghalaya

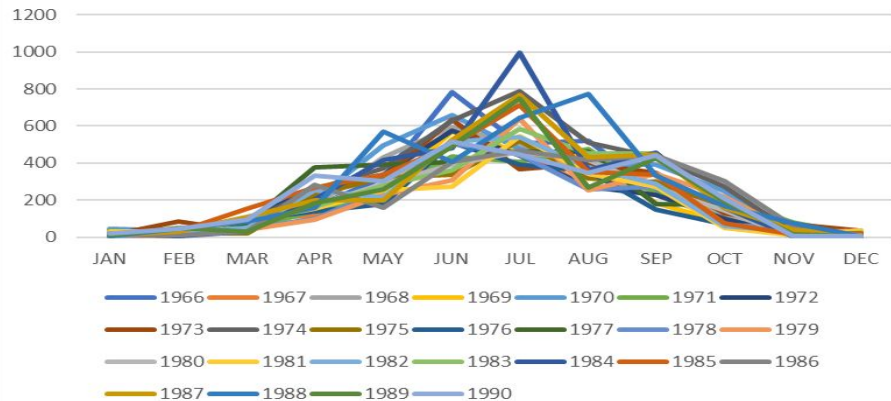
Seasonal Indices



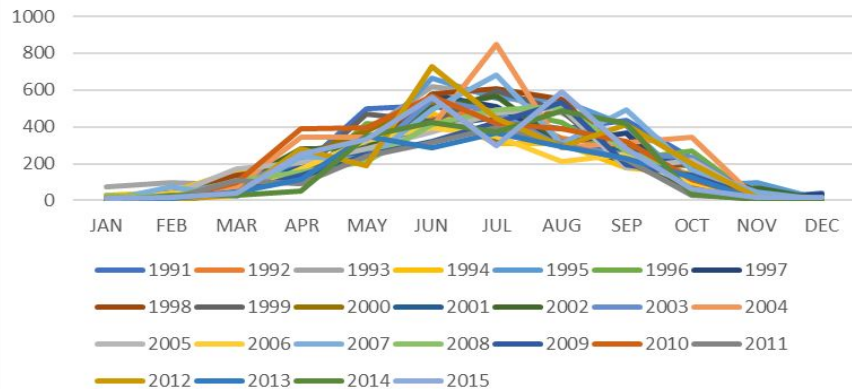
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Seasonal Component

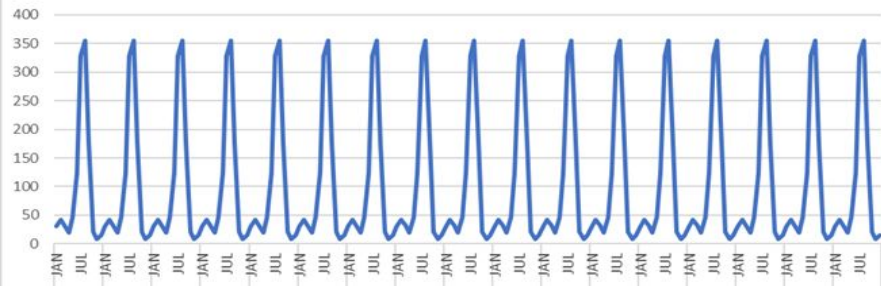


Seasonal Component

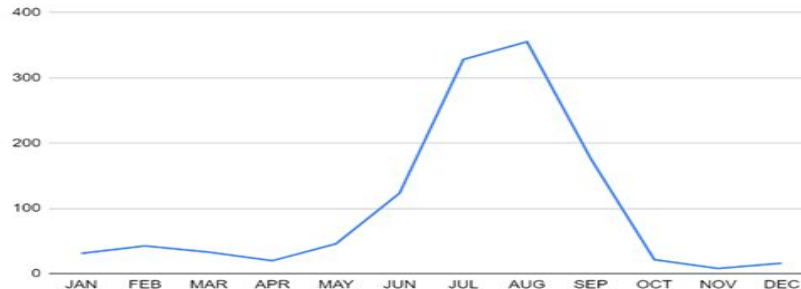


SEASONAL COMPONENT-Delhi Haryana and Chandigarh

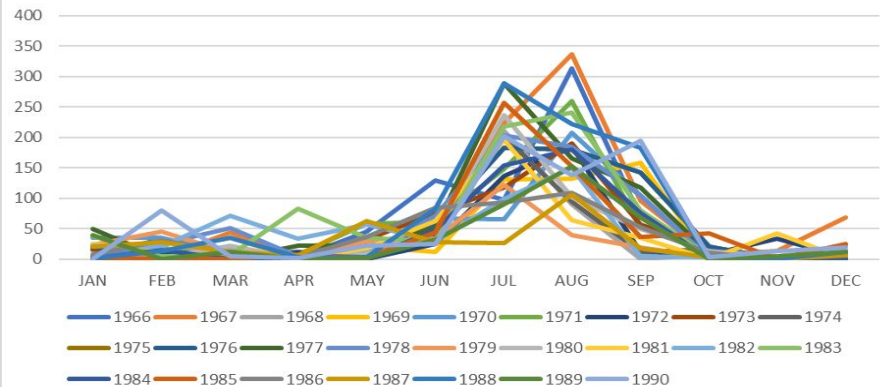
Seasonal Indices



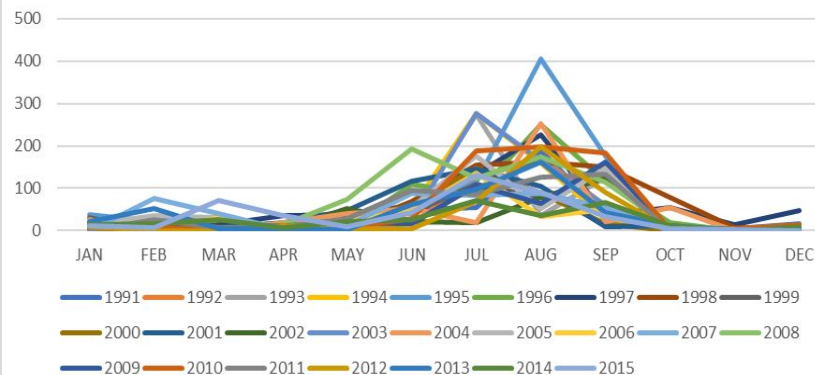
Seasonal indices



Seasonal Component



Seasonal Component



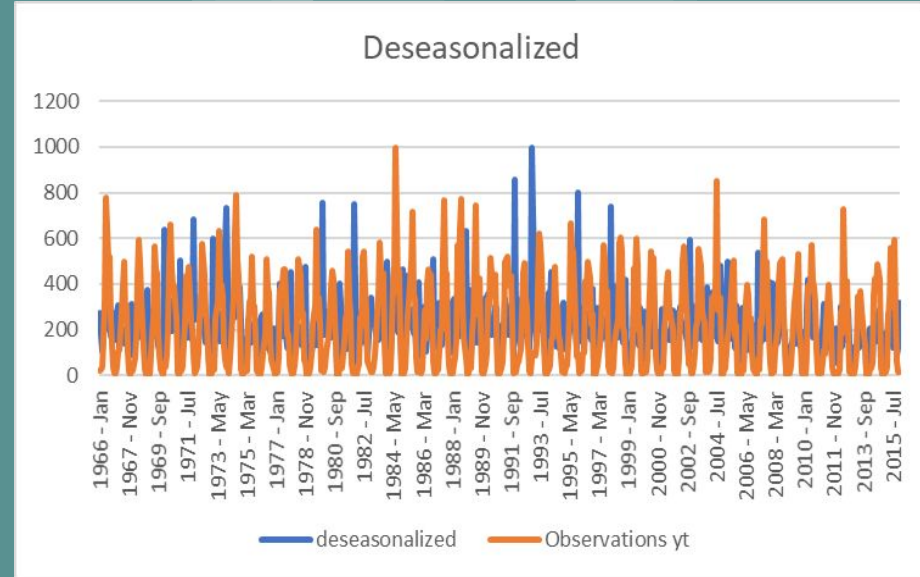
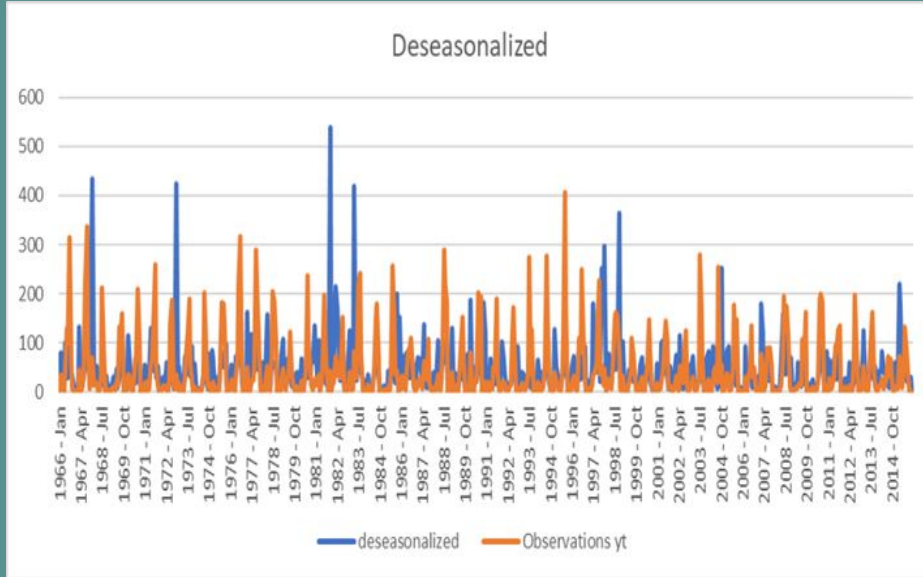
DELHI – HARYANA VS ASSAM - MEGHALAYA

BASIS	DELHI , HARYANA, CHANDIGARH	ASSAM – MEGHALAYA
Duration	June to October (4-5 months) [Due to monsoon season]	February to November (10-11 months)[Due to presence of plateau and extremely steep slopes]
Peak	July – August	June – July
Maximum rainfall recorded (in cm)	405.3	995.2
Distribution over the year	Receives large rainfall for few months	Uniform pattern of rainfall over the year
Pattern over the years	Decline	Minimal decline
Reasons/Causes	Monsoon rainfall due to Bay of Bengal branch of monsoon and winter rainfall due to Western Disturbances	Winds rise up the eastern and southern slopes of the plateau, leading to heavy rainfall

DESEASONALIZED

DELHI , HARYANA AND CHANDIGARH

ASSAM - MEGHALAYA

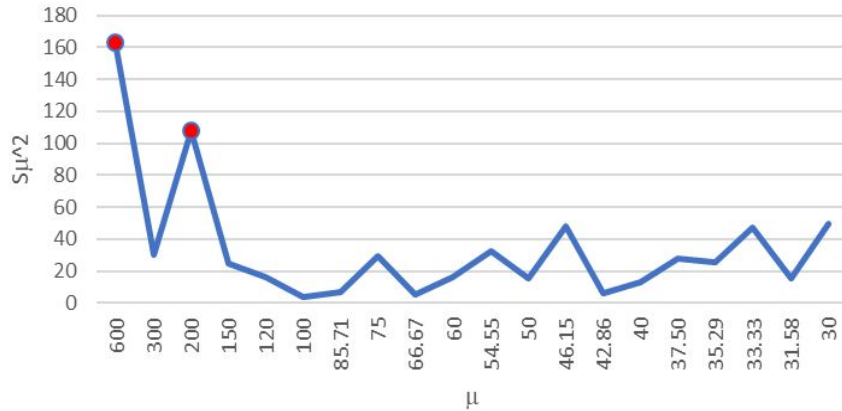


Here, the deseasonalized graphs show the remaining oscillatory and random components where as original series shows a regular periodicity which is due to seasonal component. So, we can conclude that the Seasonal Component is dominating component of the series. We can also observe that there is a variation between the original and deseasonalized time series which means other components are also playing crucial role in the time series..

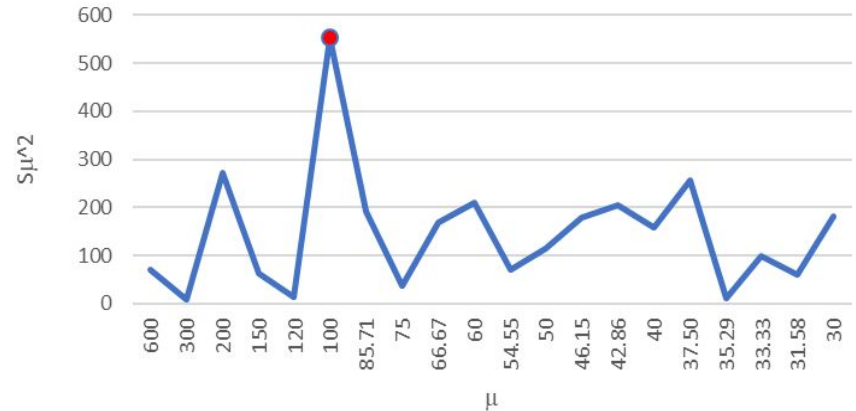
CYCLIC COMPONENT

For computing cyclic component we have used Harmonic analysis method, as per assumption of cyclic component we have removed trend and seasonal component from the time series. We have taken 20 trial periods and plotted a periodogram and with help of ANOVA we have found that for Assam & Meghalaya there are two significant trial periods and for Delhi Haryana & Chandigarh there is only one significant trial period that can be chosen as lambda that is periodicity.

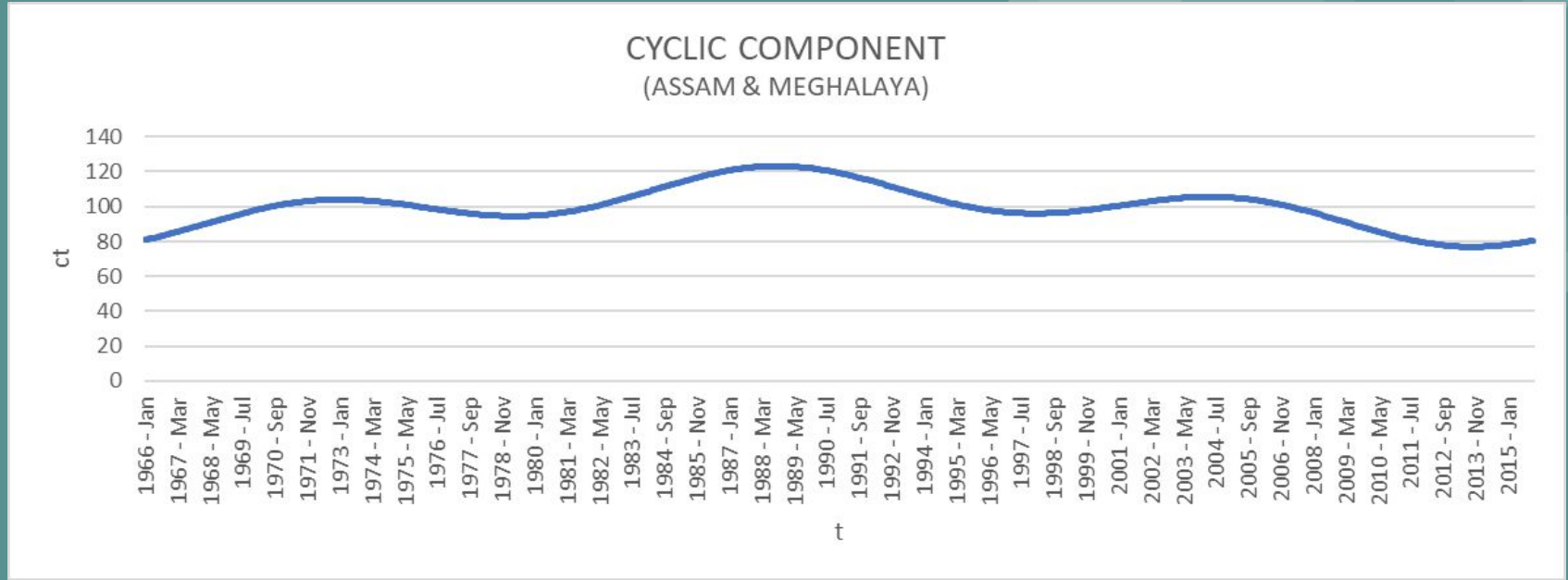
PERIODOGRAM
(ASSAM & MEGHALAYA)



PERIODOGRAM
(DELHI HARYANA & CHANDIGARH)

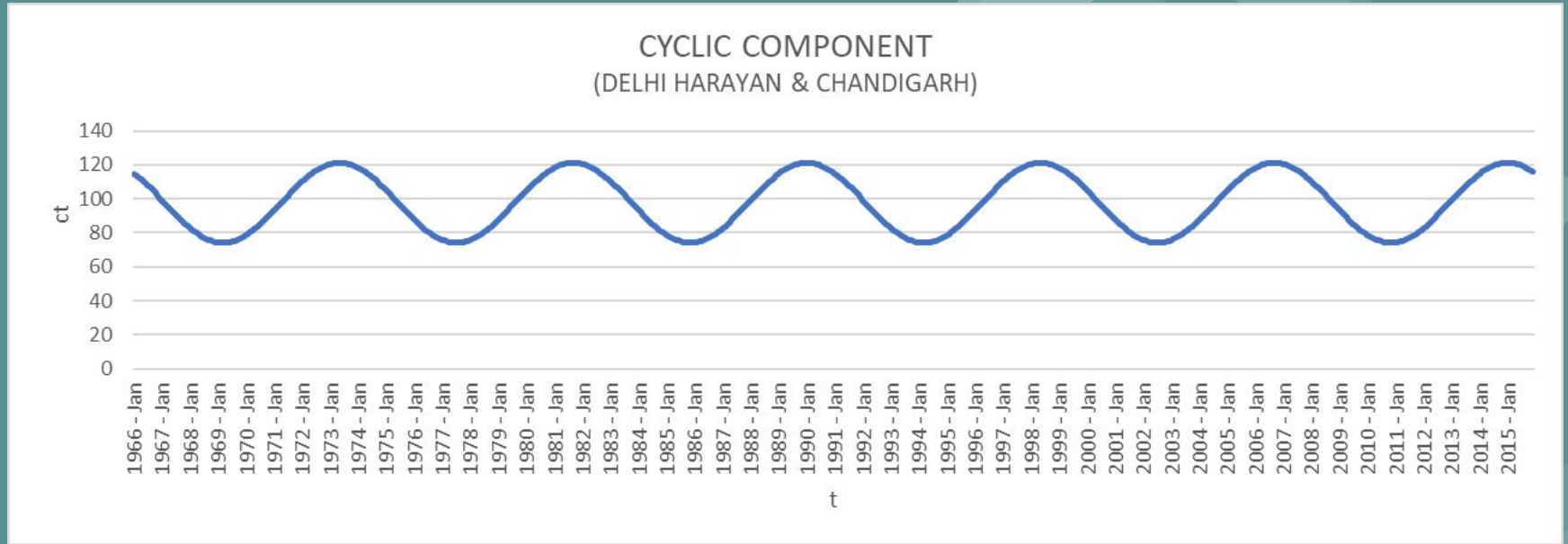


CYCLIC COMPONENT



Cyclic component can be noted significantly as we have analysed data for 50 years. The rainfall varies from 80 -120 cms gradually with moderate fluctuations over the 50 years. Reason behind this can be asserted to temperature variations such as el-nino and la-nina, change in wind patterns, atmospheric pressures and air masses.

CYCLIC COMPONENT



Cyclic component can be noted significantly as we have analysed data for 50 years. The rainfall varies from 70 -130 cms with regular fluctuations over the 50 years in this region. Reasons behind this can be asserted to change in wind patterns over the years such as western disturbances and south west monsoon winds.

VARIATE DIFFERENCE METHOD

Years	Annual average	1st diff	2nd diff
1966	232.4417		
1967	176.9417	-55.5000	
1968	210.9000	33.9583	89.4583
1969	194.3583	-16.5417	-50.5000
1970	254.0417	59.6833	76.2250
1971	192.9250	-61.1167	-120.8000
1972	198.2583	5.3333	66.4500
1973	222.6750	24.4167	19.0833
1974	283.6333	60.9583	36.5417
1975	189.2167	-94.4167	-155.3750
1976	163.0000	-26.2167	68.2000
1977	218.0583	55.0583	81.2750
1978	167.6500	-50.4083	-105.4667
1979	182.3667	14.7167	65.1250
1980	190.7667	8.4000	-6.3167
1981	173.2750	-17.4917	-25.8917
1982	198.5250	25.2500	42.7417
1983	218.3500	19.8250	-5.4250
1984	263.0750	44.7250	24.9000
1985	233.4750	-29.6000	-74.3250
1986	216.0583	-17.4167	12.1833
1987	243.4667	27.4083	44.8250
1988	271.8083	28.3417	0.9333
1989	223.5250	-48.2833	-76.6250
1990	230.9333	7.4083	55.6917
1991	236.5000	5.5667	-1.8417

1992	188.3083	-48.1917	-53.7583
1993	239.9333	51.6250	99.8167
1994	176.7750	-63.1583	-114.7833
1995	240.5000	63.7250	126.8833
1996	205.5667	-34.9333	-98.6583
1997	197.6833	-7.8833	27.0500
1998	233.5917	35.9083	43.7917
1999	216.1750	-17.4167	-53.3250
2000	207.2333	-8.9417	8.4750
2001	181.5500	-25.6833	-16.7417
2002	199.3833	17.8333	43.5167
2003	202.8500	3.4667	-14.3667
2004	249.5167	46.6667	43.2000
2005	205.6833	-43.8333	-90.5000
2006	158.2250	-47.4583	-3.6250
2007	229.3917	71.1667	118.6250
2008	194.9667	-34.4250	-105.5917
2009	172.3500	-22.6167	11.8083
2010	225.9500	53.6000	76.2167
2011	145.2917	-80.6583	-134.2583
2012	217.4500	72.1583	152.8167
2013	151.3750	-66.0750	-138.2333
2014	183.8500	32.4750	98.5500
2015	205.9167	22.0667	-10.4083
Second moment	44326.3805	1850.6600	5813.1807
Est of var	44326.3805	925.3300	968.8635
R	6.7873	-0.6532	

For Assam and Meghalaya
Clearly, $|R^2| < 1.96$.

Hence variance of the random component is σ^2_1 or σ^2_2

Also the appropriate degree of the polynomial to represent the systematic part of the time series is $k-1=0$.

So, according to this there no trend present in the time series, but as we have observed a little downfall of trend in 50 years on plotting it so we have fitted the linear trend.

VARIATE DIFFERENCE METHOD

YEAR	Annual average	1st diff	2nd diff
1966	58.9917		
1967	69.9917	11.0000	
1968	32.3583	-37.6333	-48.6333
1969	39.7667	7.4083	45.0417
1970	44.8500	5.0833	-2.3250
1971	53.0667	8.2167	3.1333
1972	36.7333	-16.3333	-24.5500
1973	43.9250	7.1917	23.5250
1974	32.0750	-11.8500	-19.0417
1975	51.8333	19.7583	31.6083
1976	60.5583	8.7250	-11.0333
1977	61.6750	1.1167	-7.6083
1978	54.2750	-7.4000	-8.5167
1979	28.4583	-25.8167	-18.4167
1980	42.7750	14.3167	40.1333
1981	41.1083	-1.6667	-15.9833
1982	42.2167	1.1083	2.7750
1983	62.0417	19.8250	18.7167
1984	39.9333	-22.1083	-41.9333
1985	47.7000	7.7667	29.8750
1986	37.5833	-10.1167	-17.8833
1987	26.5500	-11.0333	-0.9167
1988	70.4250	43.8750	54.9083
1989	34.7083	-35.7167	-79.5917
1990	58.6000	23.8917	59.6083
1991	32.8167	-25.7833	-49.6750

1992	35.7917	2.9750	28.7583
1993	43.6417	7.8500	4.8750
1994	47.0250	3.3833	-4.4667
1995	68.2667	21.2417	17.8583
1996	53.7250	-14.5417	-35.7833
1997	55.7167	1.9917	16.5333
1998	57.1083	1.3917	-0.6000
1999	28.7083	-28.4000	-29.7917
2000	32.0333	3.3250	31.7250
2001	40.4500	8.4167	5.0917
2002	29.7750	-10.6750	-19.0917
2003	49.7833	20.0083	30.6833
2004	40.6000	-9.1833	-29.1917
2005	43.2833	2.6833	11.8667
2006	27.5833	-15.7000	-18.3833
2007	37.2167	9.6333	25.3333
2008	59.1917	21.9750	12.3417
2009	32.9417	-26.2500	-48.2250
2010	53.3667	20.4250	46.6750
2011	42.6750	-10.6917	-31.1167
2012	32.6083	-10.0667	0.6250
2013	38.4417	5.8333	15.9000
2014	25.4583	-12.9833	-18.8167
2015	36.2917	10.8333	23.8167
Second moment	44326.3805	1850.6600	5813.1807
Est of var	44326.3805	925.3300	968.8635
R	6.7873	-0.6532	

For Delhi Haryana & Chandigarh
Clearly, $|R2| < 1.96$.

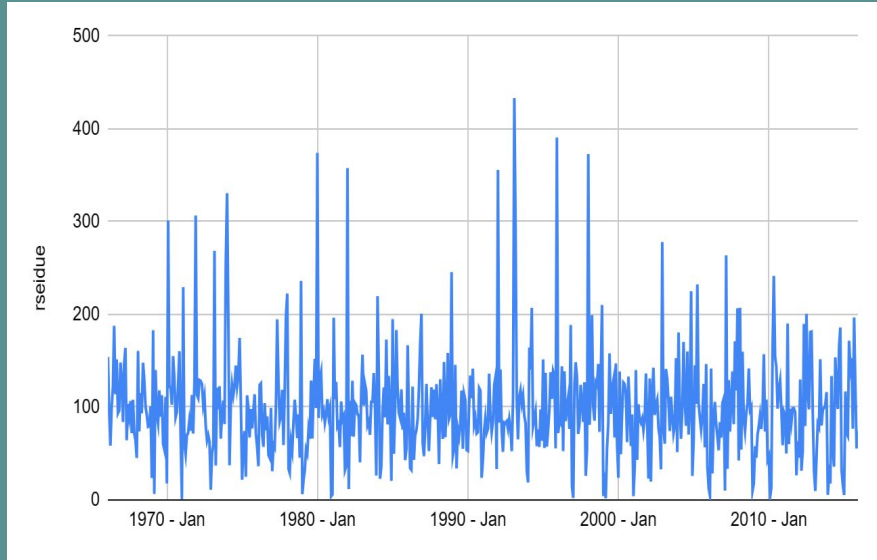
Hence variance of the random
component is σ^2_1 or σ^2_2

Also the appropriate degree of
the polynomial to represent the
systematic part of the time series
is $k-1=0$.

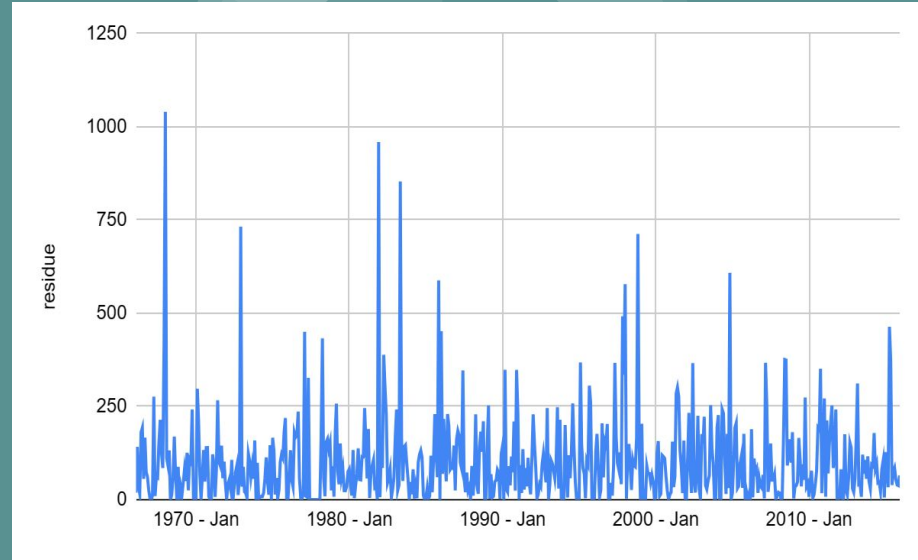
So, according to this there no
trend present in the time series,
but as we have observed a little
downfall of trend in 50 years on
plotting it so we have fitted the
linear trend.

RANDOM COMPONENT

ASSAM & MEGHALAYA



DELHI HARYANA & CHANDIGARH



Rainfall is inherently variable and influenced by a complex interplay of atmospheric, meteorological, and climatic factors. Natural variability can result in random fluctuations in the data, making it challenging to identify consistent patterns. The Random pattern of assam and meghalaya ranges between 0 to 400 whereas for delhi haryana and chandigarh it ranges between 0 to 1000 which means extreme fluctuations due to random component.

COMPARATIVE ANALYSIS AND CONCLUSION

ASSAM AND MEGHALAYA

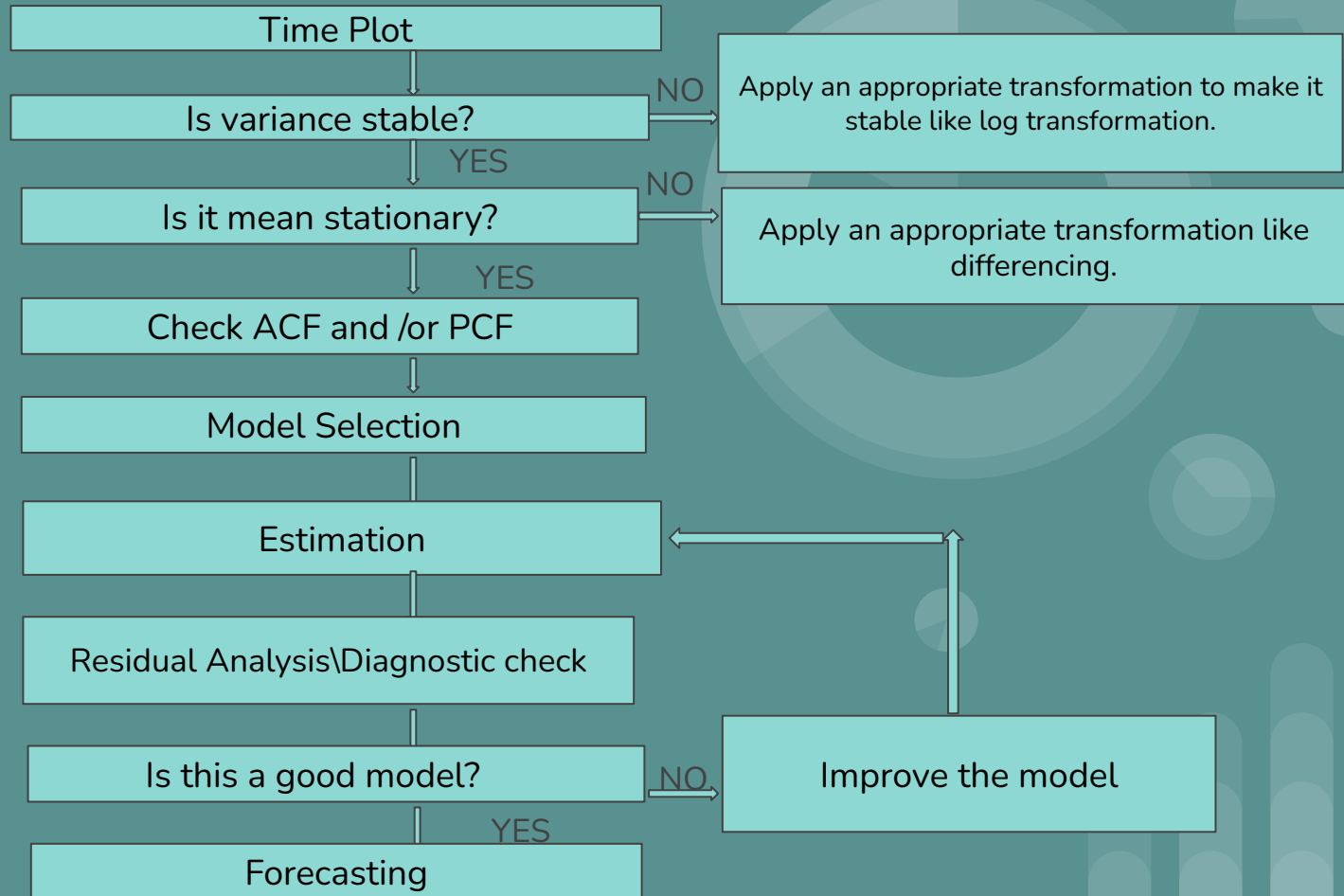
1. TREND: Over a large period of time, it only shows a slight decline in rainfall patterns.
2. SEASONAL COMPONENT:
Uniform pattern of rainfall over the year
3. CYCLIC COMPONENT: The rainfall varies gradually with moderate fluctuations over 50 years
4. RANDOM COMPONENT:
It has comparatively less Random fluctuations than delhi haryana and chandigarh region.

DELHI,HARYANA AND CHANDIGARH

1. TREND: A decline is observed in the rainfall patterns.
2. SEASONAL COMPONENT:
Receives large rainfall for few months
3. CYCLIC COMPONENT: The rainfall varies regularly with high fluctuations over 50 years
4. RANDOM COMPONENT:
It has comparatively more and extreme random fluctuations than assam and meghalaya.

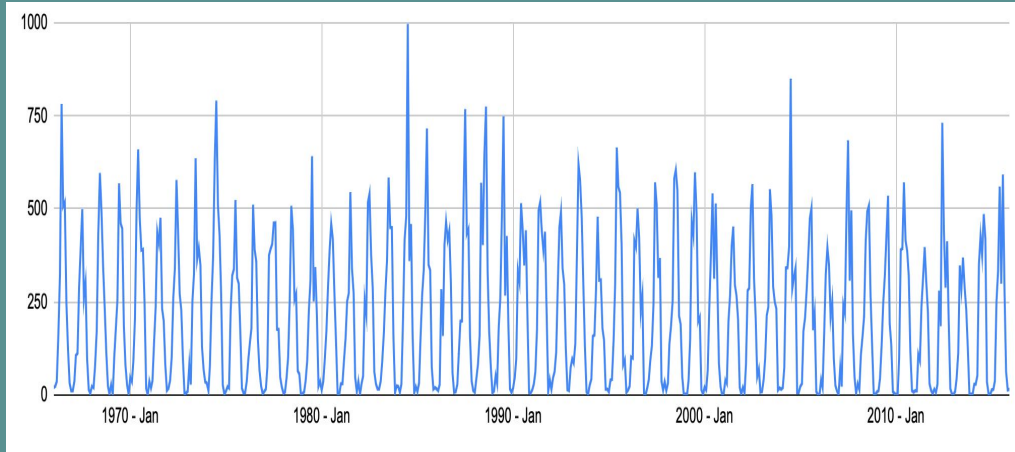
In conclusion, the rainfall data from both the regions reveals a prominent seasonal component that dominates the data, while a minimal linear trend, cyclic patterns, and random fluctuations are also present.

BOX- JENKINS FLOW CHART



TEST STATIONARITY OF THE TIME SERIES

ASSAM AND MEGHALAYA



We can observe from the given time plot that our time series data is concentrated around a constant mean value, hence it is **mean stationary**.

Also, all the values are lying within a range and there is almost no trend present in the data and therefore it is **variance stationary**. Hence, the given time series is **STATIONARY**.

The Augmented Dickey-Fuller Test (adf-test)

The null hypothesis

H0 : The time series is non-stationary

The alternative hypothesis

H1: The time series is stationary

Result:

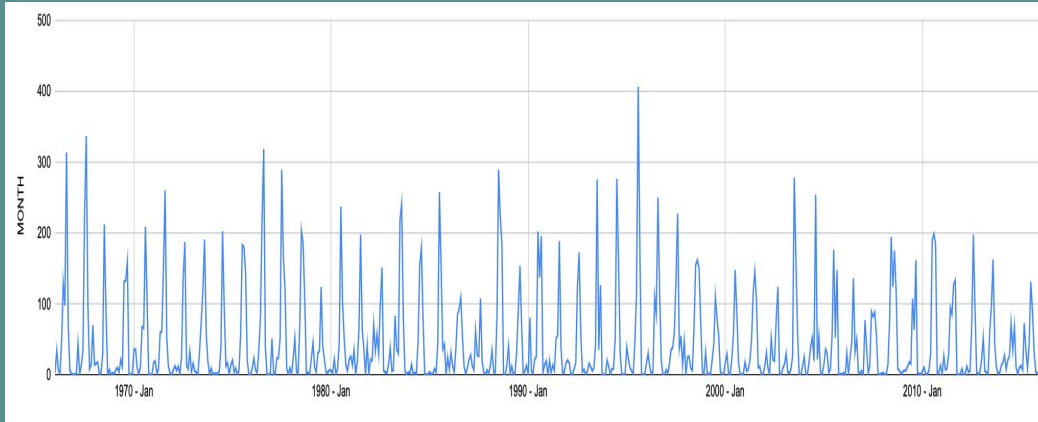
Dickey-Fuller = -13.321, Lag order = 8, p-value = 0.01

Since the p-value $0.01 < 0.05$, we reject the null hypothesis and conclude that the given time series is stationary.

By both visualization and ADF test, we can conclude that the given time series is STATIONARY

TEST STATIONARITY OF THE TIME SERIES

DELHI, HARYANA AND CHANDIGARH



We can observe from the given time plot that our time series data is concentrated around a constant mean value, hence it is **mean stationary**. Also, all the values are lying within a range and there is almost no trend present in the data and therefore it is **variance stationary**. Hence, the given time series is **STATIONARY**.

The Augmented Dickey-Fuller Test (adf-test)

The null hypothesis

H0 : The time series is non-stationary

The alternative hypothesis

H1: that the time series is stationary

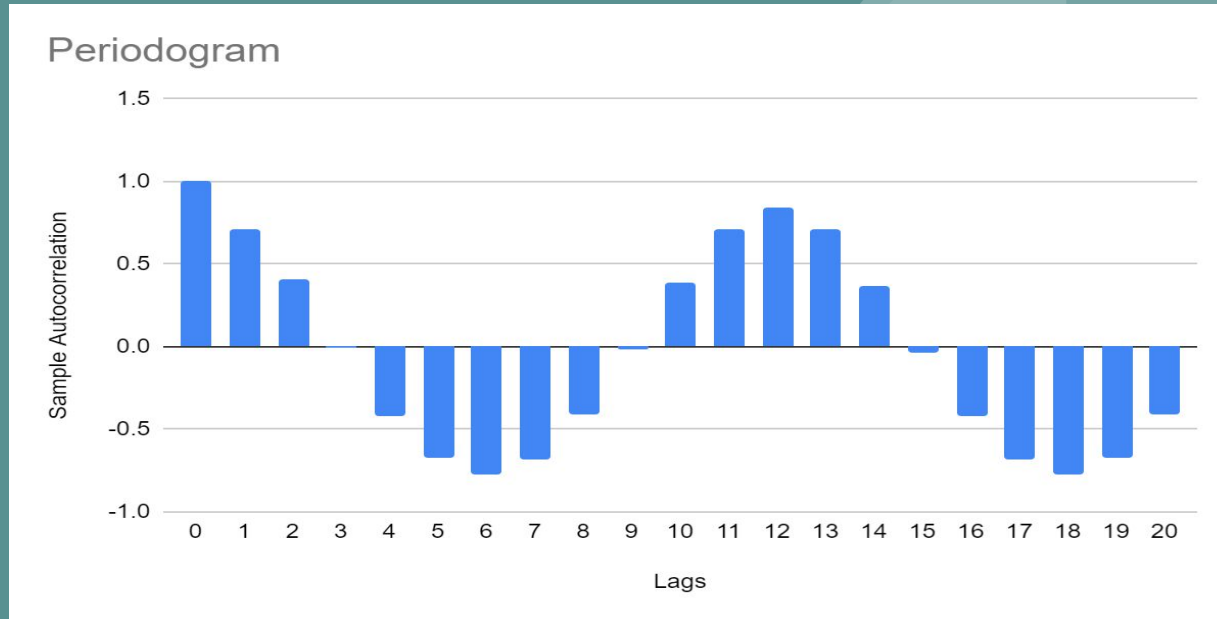
Result:

Dickey-Fuller = -17.806, Lag order = 8, p-value = 0.01

Since the p-value, $0.01 < 0.05$, we reject the null hypothesis and conclude that the given time series is stationary.

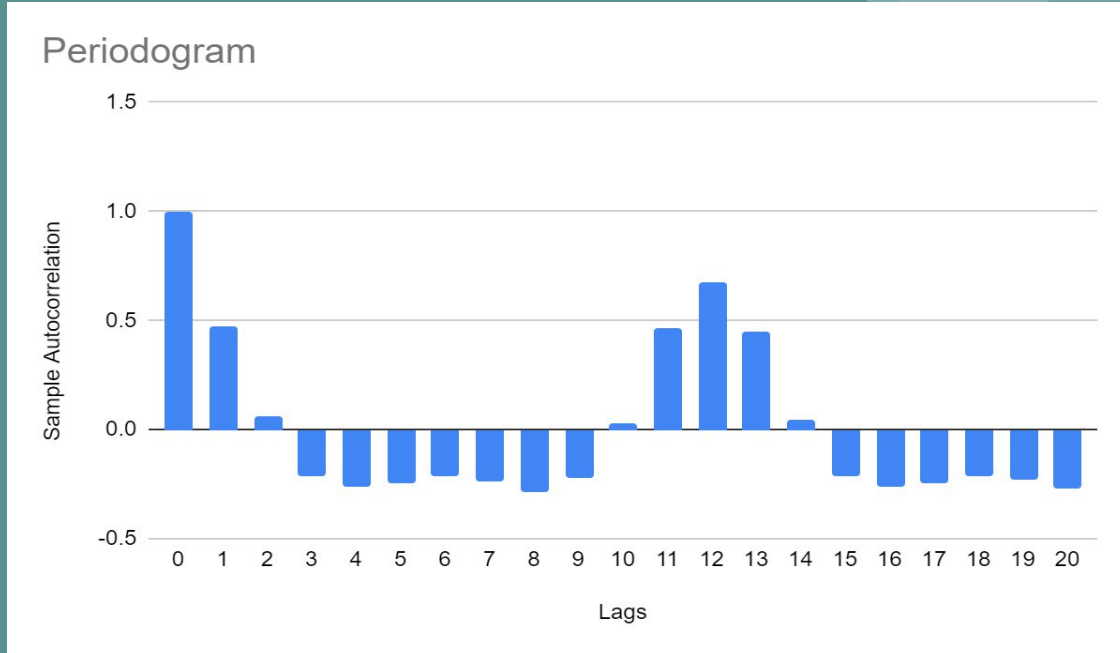
By both visualization and ADF test, we can conclude that the given time series is STATIONARY

CORRELOGRAM OF ASSAM AND MEGHALAYA



We observe periodic spikes at regular intervals. Thus, there's a clear seasonality in the data. Since, there's a spike at lag 12, it suggests a yearly seasonality in the data. Negative autocorrelation at certain lags suggests an inverse relationship, that is, there's a pattern of alternating highs and lows with a periodicity of 6 time units.

CORRELOGRAM OF DELHI, HARYANA AND CHANDIGARH



We observe periodic spikes at regular intervals. Thus, there's a clear seasonality in the data. Since, there's a spike at lag 12, it suggest a yearly seasonality in the data. A positive autocorrelation at a certain lag suggests that high values at the current time point are associated with high values at that lagged point in the past, and vice versa. Negative autocorrelation at certain lags suggests an inverse relationship.

FITTING A TIME SERIES MODEL

ASSAM AND MEGHALAYA

Using SPSS and R, we have fitted three possible models and performed an analysis for finding the best possible model and hence calculate the forecasts.

PROPERTIES	WINTER'S - MULTIPLICATIVE MODEL	SARIMA(0,0,3)(0,1,1) [12]	ARIMA(5,0,1) [as d=0, it is ARMA(5,1)]
DRIFT	NO DRIFT	NO DRIFT	DRIFT IS PRESENT, mu = 208.38
EQUATION	$\hat{y}_{t+h} = (a + h \cdot g) \cdot s_t \cdot x_t^{((h-1)/m)}$, where a(level)=0.006, g(trend)=0.000, d(seasonal)=0.008	$(1 - 0 \cdot B + 0.101B^2 + 0.139B^3)(1 - 0.961B^{12})(1 - B^{12})y_t = \epsilon_t$	$(1 - 0.93B + 0.61B^2 + 0.135B^3 + 0.244B^4 + 0.07B^5)(1 - 0.564B)y_t = \epsilon_t + 208.38$
R SQUARE	0.841	0.829	0.733
RMSE	78.405	81.073	101.998
MAE	52.496	54.324	75.907

FITTING A TIME SERIES MODEL DELHI, HARYANA AND CHANDIGARH

Using SPSS and R, we have fitted three possible models and performed an analysis for finding the best possible model and hence calculate the forecasts.

PROPERTIES	SIMPLE SEASONAL MODEL	SARIMA(0,0,0)(0,1,1) [12]	ARIMA(2,0,2) [as d=0, it is ARMA(2,2)]
DRIFT	NO DRIFT	NO DRIFT	DRIFT IS PRESENT, mu = 44.3917
EQUATION	$\hat{y}_{t+h} = s(t-m+h)$, Where, a(level)=0.006,g(trend)=0.000, d(seasonal)=8.231E-07	$(1-0.9251B^{12})(1-B^{12})y_t = \epsilon_t$	$(1-1.36B+0.653B^2)(1-0.939B+0.120B^2)y_t = \epsilon_t + 44.3917$
R SQUARE	0.657	0.616	0.333
RMSE	38.260	40.199	53.510
MAE	23.148	24.084	36.747

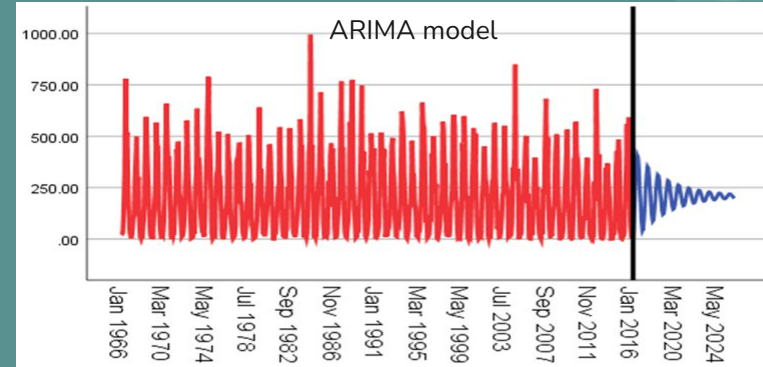
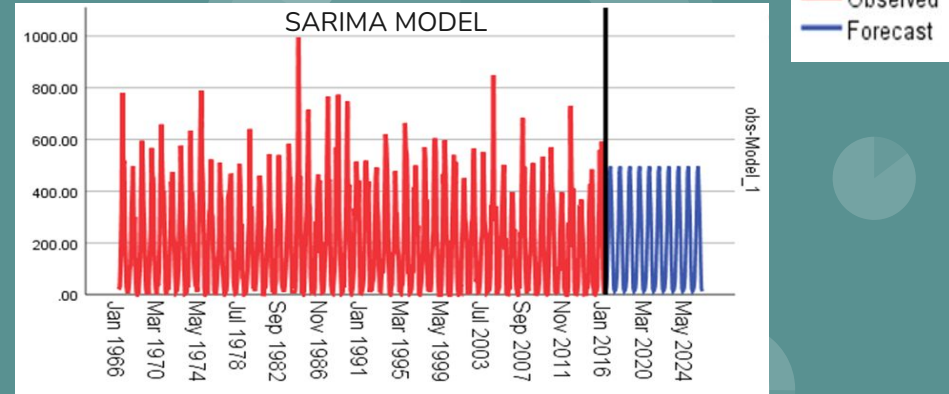
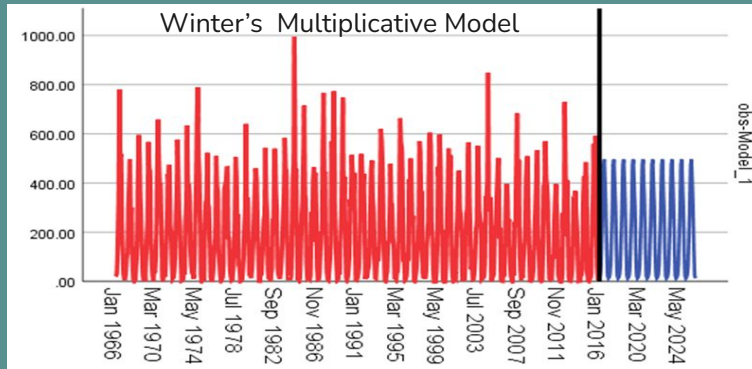
SELECTION OF BEST FITTED MODEL

ASSAM AND MEGHALAYA

Among all three models,

1. Maximum value of R Square is coming out for Winter's Multiplicative model.
2. Minimum value of RMSE is coming out for Winter's Multiplicative model.
3. Minimum value of MAE is coming out for Winter's Multiplicative model.

Hence, we can conclude that **Winter's Multiplicative** is the best fit model for the given time series data.



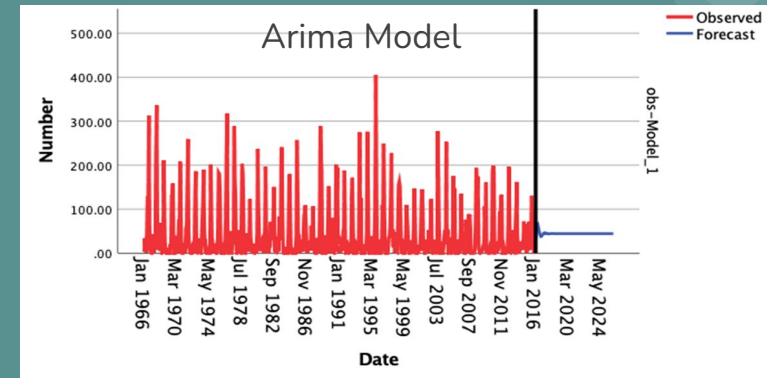
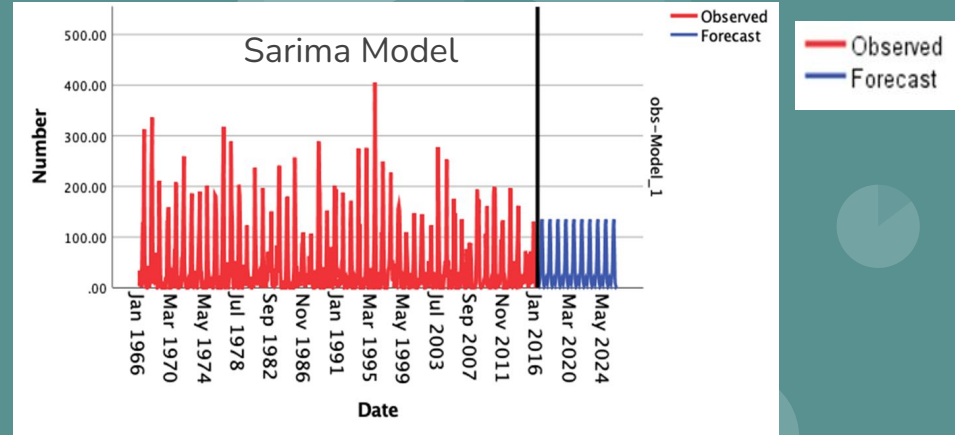
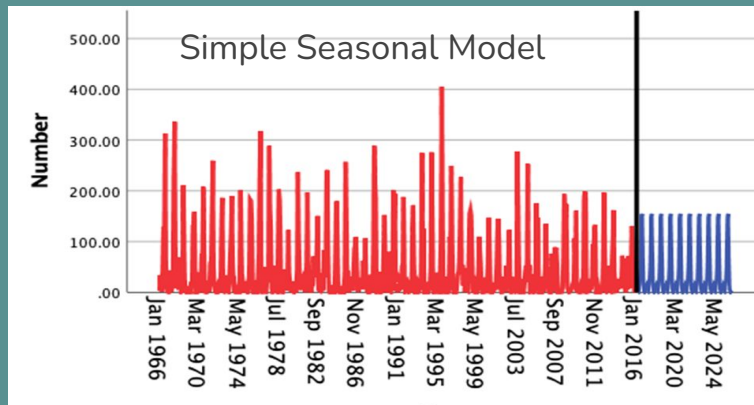
SELECTION OF BEST FITTED MODEL

DELHI, HARYANA AND CHANDIGARH

Among all three models,

1. Maximum value of R Square is coming out for Simple Seasonal model.
2. Minimum value of RMSE is coming out for Simple Seasonal model.
3. Minimum value of MAE is coming out for Simple Seasonal model.

Hence, we can conclude that **Simple Seasonal** is the best fit model for the given time series data.

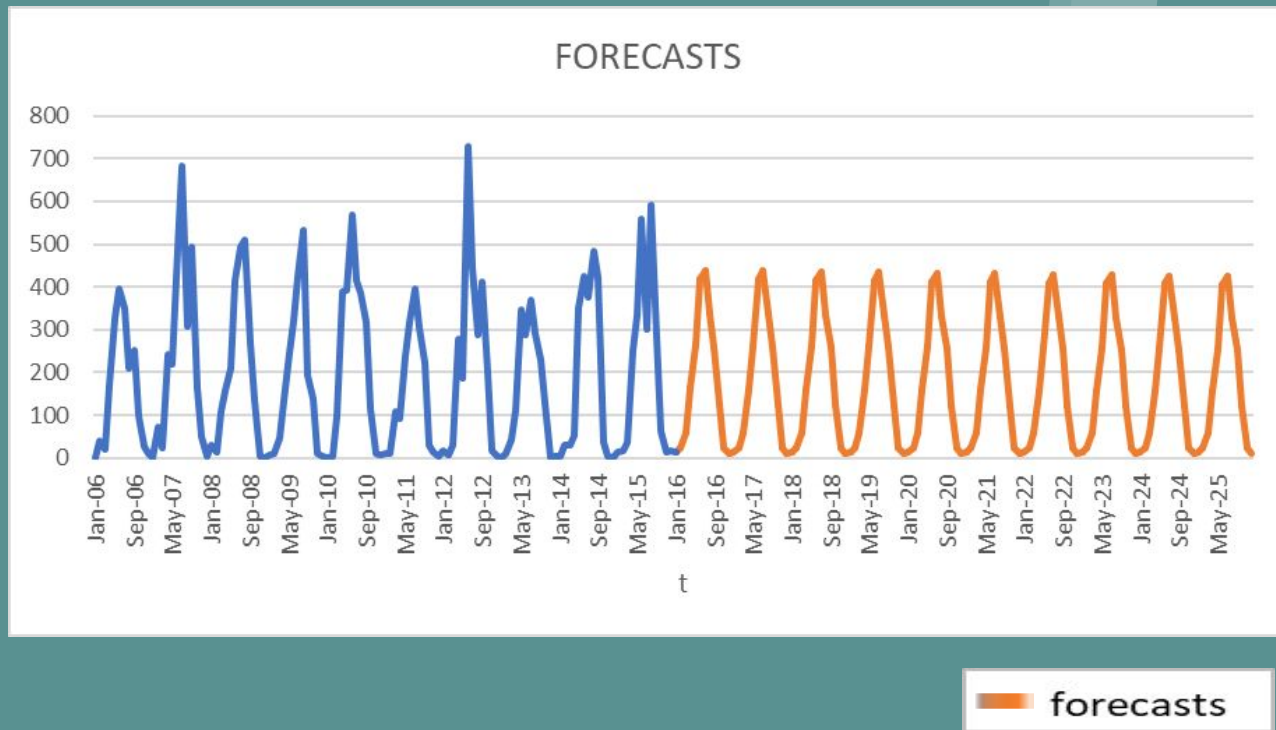


FORECASTS - ASSAM AND MEGHALAYA

By Winter's Multiplication model

FORECAST FOR 10 YEARS

Plot from 2006 to 2025, 10 years observations and 10 forecasted years for better visualisation.



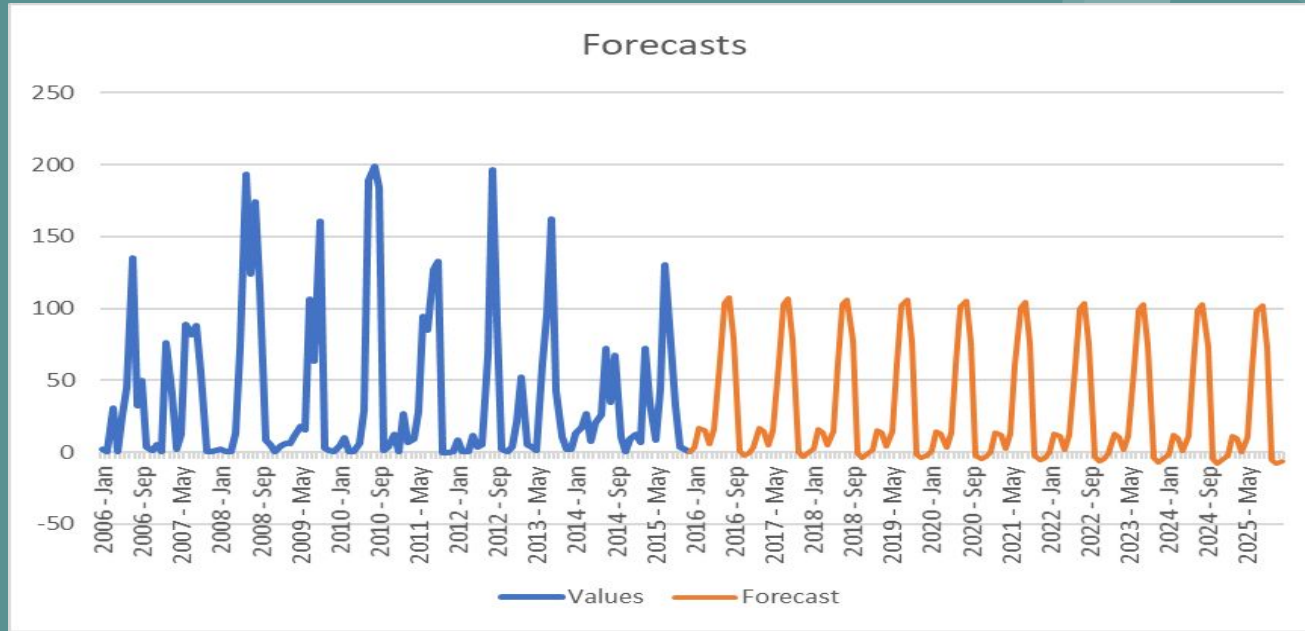
We can see that the forecasted values are showing regular fluctuations and we know that in our time series data seasonal component is dominating. Hence, it is forecasting majorly the seasonal component. As the Winters Multiplicative Model is particularly effective for time series data with a strong seasonal pattern, and it allows for capturing both short-term fluctuations and long-term trends in the data

FORECASTS - DELHI AND HARYANA

By Simple Seasonal model

FORECAST FOR 10 YEARS

Plot from 2006 to 2025, 10 years observations and 10 forecasted years for better visualisation.



Forecasted values are showing regular fluctuations and we know that in our time series data seasonal component is dominating. Hence, it is forecasting majorly the seasonal component. A simple seasonal model is a time series forecasting model that takes into account the seasonal patterns in the data.

forecasts

COMPARATIVE ANALYSIS AND CONCLUSION

ASSAM AND MEGHALAYA

1. We observed that the Winter's Multiplicative Model is suitable for forecasting our chosen data set
2. The forecasting shows seasonal pattern. Periodic spikes at regular intervals are observed.
3. Correlogram shows a seasonal pattern which indicates its periodicity i.e., 6. After every 6 months opposite pattern is observed.

DELHI, HARYANA AND CHANDIGARH

1. Here, the Simple Seasonal Model is used for forecasting data.
2. Forecasting shows majorly seasonal pattern.
3. Correlogram shows seasonal pattern but here periodicity is 1 year i.e., it repeats itself after an year.

CONCLUSION

1. We have used Multiplicative Model whose equation is $Y_t = T_t S_t R_t C_t$
2. Our chosen dataset contains all the four components trend, Seasonal, Cyclic and Random.
3. Based on the all the calculations carried out , we observed that Seasonal component dominates the rainfall pattern. Trend has minimal downward movement over the years and Cyclic and Random factors influence rainfall patterns moderately.
4. We can observe from the correlogram as well that the Seasonal Component is prominent in rainfall time series data.
5. It can be seen that for Delhi & Haryana Simple Seasonal Model is the best fitted Model for forecasting future values whereas for Assam & Meghalaya Winter's Multiplicative Model is best for prediction of pattern.